Comparing Structural and Reduced-Form Credit Risk Models for Default Risk Measurement: Merton, VK, and Hull-White

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1. Introduction

Credit risk modelling falls into two broad families (Duffie & Singleton, 2003). Structural models — pioneered by Merton (1974) and refined by the Vasicek-Kealhofer (VK/KMV) framework of Crosbie and Bohn (2003) — derive default endogenously from the firm's asset value process: equity is treated as a European call on assets, and default is triggered when assets fall below a debt-related barrier. Reduced-form (intensity) models, such as Hull and White (2000), bypass the unobservable asset value and model the default event directly through a hazard rate $\lambda(t)$.

This study addresses a single research question: which paradigm — structural or reduced-form — provides superior default discrimination for U.S.-listed firms? We implement three representative models (Merton, VK/KMV, Hull-White) on a common firm-year panel spanning 2021–2025 and evaluate them with the receiver operating characteristic area under the curve (ROC-AUC) and the cumulative accuracy profile accuracy ratio (CAP-AR), the two standard discrimination metrics in the credit risk literature (Engelmann, Hayden, & Tasche, 2003). The reduced-form specification follows the discrete-time hazard approach of Shumway (2001)

2. Data

2.1 Sources and Sample

The baseline panel covers **163 U.S.-listed firms** observed annually from **2021 to 2025**, yielding **659 firm-year observations**. Firms are drawn from a universe of large- and mid-capitalisation names spanning financials, industrials, energy, and technology sectors. Annual balance-sheet data—total debt, short-term debt (STD), long-term debt (LTD), and total assets—are sourced from Yahoo Finance. Risk-free rates are taken from the Federal Reserve Economic Data (FRED) database using the U.S. Treasury constant-maturity series. Implied volatility is sourced from OptionMetrics Ivy DB US (WRDS) (Appendix 2.1.1).

2.2 Implied Volatility: Data Source & Filtering

In the Merton (1974) framework, equity volatility is the observable input used to identify the unobservable pair (V, Sigma_V). The conventional estimator is purely backward-looking and may be a poor proxy for default risk over the next year. We instead use option-implied volatility, which embeds the market's forward-looking expectation of equity volatility over the contract's horizon and is therefore the more natural input for a model that prices forward default risk (Hull, 2018; Bharath & Shumway, 2008). For each firm-year, we extract a single at-the-money (ATM), approximately one-year call-option implied volatility from OptionMetrics, applying the filters: year-end window, days-to-expiry $\in [0.40, 0.60]$, $\Delta \in [0.40, 0.60]$, $0 < \text{IV} < 3$, positive volume or open interest, and one observation per firm-year selected by Δ closest to 0.50 (ties broken by tenor closest to 365 days). When option data are unavailable (mostly 2025 delisted distressed firms), σ_E

falls back to $\sigma_{hist,E}$ on a rolling 252-day window, ensuring full panel coverage (Appendix 2.2.1, Appendix 2.2.2).

2.3 Default Label: Broad Credit-Distress Definition

Because formal Chapter 11 filings among U.S.-listed firms during 2021–2025 are rare, restricting the positive class to legal bankruptcy would yield severe class imbalance and an unstable hazard estimator (Shumway, 2001; Campbell, Hilscher, & Szilagyi, 2008). We therefore adopt a broad credit-distress label, assigned at the firm level (static across years) and defined as the union of three tiers. Applying this methodology yields a total of 21 default firms.

$$y_i = \mathbf{1}\{T_i^{(1)} \vee T_i^{(2)} \vee T_i^{(3)}\}, \quad y_i \in \{0, 1\}.$$

The three tier indicators are:

$$T_i^{(1)} = \mathbf{1}\{\text{firm } i \text{ has filed for Chapter 11 in the sample window}\},$$

$$T_i^{(2)} = \mathbf{1}\{\text{auditor/10-K issues a substantial-doubt going-concern opinion}\},$$

$$T_i^{(3)} = \mathbf{1}\{C_1 \vee C_2 \vee C_3 \vee C_4\}.$$

Tier-3 (severe financial distress) triggers if any of the following four firm-year conditions are observed at least once during 2021–2025:

$$C_1 : NI_{i,t} < 0 \text{ for } \geq 3 \text{ consecutive years (persistent net losses)},$$

$$C_2 : FCF_{i,t} < 0 \text{ and } Cash_{i,t}/|FCF_{i,t}| < 2 \text{ (unsustainable cash burn)},$$

$$C_3 : BE_{i,t} < 0 \text{ (negative book equity)},$$

$$C_4 : D_{i,t}/D_{i,t} + E_{i,t} > 0.90 \text{ (extreme leverage)}.$$

3. Models

This section introduces the three credit risk models used in our analysis: the Merton (1974) structural baseline, the Vasicek–Kealhofer / KMV refinement (Crosbie & Bohn, 2003), and the CDS-based Hull–White (2000) reduced-form hazard rate.

3.1 Merton (1974)

The Merton model treats equity as a European call option on the firm's asset value *V* struck at the face value of debt *D*, maturing at horizon *T*. Under the standard assumptions of geometric Brownian motion for *V* and a single zero-coupon debt claim, equity value and equity volatility satisfy the simultaneous system

$$E = VN(d_1) - De^{-rT}N(d_2)$$

$$\sigma_E = VN(d_1)\sigma_V$$

which we solve jointly for the latent pair (V, σ_V) given observable E and σ_E. The distance to default and physical-measure default probability follow as

$$DD = \frac{\ln\left(\frac{V}{D}\right) + \left(r - \frac{1}{2}\sigma_V^2\right)}{\sigma_V\sqrt{T}}, PD = N(-DD)$$

Equity volatility is sourced from OptionMetrics ATM implied volatility where available (delta ≈ 0.5, approximately 1-year maturity), providing a forward-looking estimate consistent with the option-pricing foundation of the model. For firm-years where implied volatility is unavailable, 252-day historical volatility serves as a fallback.

3.2 Vasicek–Kealhofer / KMV (Crosbie & Bohn, 2003)

KMV preserves the Merton option-pricing skeleton but introduces two practical refinements. First, the default barrier is redefined as $DP = STD + 0.5 \cdot LTD$, recognizing that firms typically default well before total liabilities are exhausted because long-term debt does not come due immediately. Second, (V, σ_V) is recovered through an iterative fixed-point scheme

$$V^{(k+1)} = \frac{E + DPe^{-rT}N(d_2^{(k)})}{N(d_1^{(k)})}$$

iterated until $|\Delta V| < 10^{-6}$, which is numerically more stable than the simultaneous Merton system. The risk-neutral drift r is replaced by an asset drift $\mu = r + 0.02$ to approximate an empirical equity premium, giving

$$DD_{VK} = \frac{\ln\left(\frac{V}{DP}\right) + \left(\mu - \frac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}}, \mu = r + 0.02$$

3.3 Hull–White

As a reduced-form benchmark, we follow Hull–White (2000) and bootstrap a flat hazard rate from the CDS pricing identity, which is consistent with Markit quotes by construction. The market-implied benchmark uses the Markit 5Y default probability dp directly:

$$\lambda_{market} = -\frac{\ln(1 - dp)}{5}, \quad PD_{1Y}^{market} = 1 - exp(-\lambda_{market})$$

which we solve via Brent's method on the quarterly payment schedule. Rather than applying a uniform recovery rate, we estimate a firm-specific RR following Altman & Kishore (1996):

RR = (STD / D) × 0.55 + (LTD / D) × 0.35where short-term debt (typically secured) is assigned 55% and long-term debt (typically unsecured) 35%. The sample mean is 0.356 with a standard deviation of 0.049, ranging from 0.20 to 0.55. Firm-years without balance-sheet decomposition fall back to the ISDA convention of 0.40.

A useful closed-form approximation is the credit-triangle relation

$$\widehat{PD}_T^{HW} = \sigma(\beta_0 + \beta_1 Lev + \beta_2 \sigma_E + \beta_3 logE)$$

$$\hat{\lambda}_{HW} = -\frac{\ln(1 - \widehat{PD}_T^{HW})}{T}$$

where *s* is the par CDS spread.

Together, these three models span the methodological spectrum: Merton as the canonical structural baseline, VK/KMV as its industry-standard refinement, and the CDS-bootstrapped Hull–White hazard as a reduced-form benchmark grounded directly in observed credit spreads.

4.Result

4.1 Merton

The Merton results show strong top-end default-risk discrimination. Among the top-15 observations by Merton PD, 14 are distressed or default-labelled firms, indicating that the model is effective at identifying severe structural credit risk. The only exception is SPCE 2024, which is not labelled as default but receives a high PD; this is better interpreted as a high-risk non-default or early-warning case rather than a simple model error. The strongest signal is BYND 2025, whose PD reaches 61.86%, far above the second-ranked SKLZ 2022 at 34.90%. This gap is economically meaningful because BYND is the only negative-DD observation, with DD = -0.3018, implying that the estimated asset value falls below the debt barrier. The repeated appearance of firms such as OPEN, WKHS, BYND, and AMC across multiple years further suggests that Merton is capturing persistent balance-sheet distress rather than one-year noise. Overall, Merton provides a strong structural baseline, although its use of total debt as the default barrier may overstate short-term default risk (Appendix 4.1.1).

4.2 Vasicek–Kealhofer / KMV

VK preserves Merton's strong top-end ranking but produces a more conservative version of the structural risk signal. Similar to Merton, 14 of the top-15 observations by VK PD are distressed or default-labelled firms, and the only high-risk non-default exception is again SPCE 2024. However, SPCE moves lower in the VK ranking and its PD falls from Merton's 28.56% to 19.69%, suggesting that the refined default point reduces some of Merton's aggressiveness. BYND 2025 remains the extreme tail observation, with VK PD = 51.29%, still far above the second-ranked OPEN 2022 at 29.08%. Compared with Merton, VK generally raises DD and lowers PD because

it replaces total debt with the lower default point, STD + 0.5 × LTD. As a result, the Top-15 tickers are almost unchanged relative to Merton, with only one ticker difference, but their internal ordering changes. This suggests that VK does not materially change which firms are identified as distressed; instead, it adjusts the intensity and relative ordering of structural risk (Appendix 4.2.1).

4.3 Hull-White

Hull-White provides a CDS-implied market-risk view rather than a structural distress ranking. The model bootstraps an annual hazard rate from 5-year CDS spreads and then converts it into 1-year and 5-year PDs. The CDS panel covers 398 firm-year observations across 84 firms from 2021 to 2025, with 5-year spreads ranging from 18 to 379 bps. Overall, the average bootstrapped hazard rate is 0.93%, corresponding to an average 1-year PD of 0.92% and an average 5-year PD of 4.48%, indicating low average annual credit risk in the CDS-covered universe (Appendix 4.3.1). However, the upper tail is much riskier: the maximum 5-year PD reaches 27.00%, and the highest-risk observations are DAL 2022, F 2022, and MU 2021, all with 5-year PDs above 20%. There is also a visible gap after the top three, with the fourth-ranked observation, DAL 2021, falling to 16.96% (Appendix 4.3.2). This top-risk list differs from Merton and VK because Hull-White is based on larger CDS-covered firms, mostly BB/BBB names in the top-risk list, while the structural models identify distress in the broader equity/default-label panel. As a sanity check, the bootstrapped 5-year PD closely reproduces Markit-implied 5-year default probabilities, with a Spearman rank correlation of 0.9987 and a mean residual of -0.001122. This confirms implementation consistency (Appendix 4.3.3).

5. Structural Discrimination (Merton vs VK)

Structural Discrimination tests whether the structural models rank distressed firms above non-default firms. Both models do this very well. Merton has an AUC of 0.9819 and CAP-AR of 0.9639, while VK has an AUC of 0.9817 and CAP-AR of 0.9634. The difference is extremely small, and the ROC curves almost overlap. This suggests that VK does not really improve the ranking performance over Merton, even though it changes the default point. The CAP curves also support the same conclusion, since most default observations are captured early among the high-PD firms rather than being randomly distributed (Appendix 5.1). The DD distribution gives the economic reason behind this performance: default observations mostly appear in the low-DD area, while non-default observations are concentrated at higher DD levels (Appendix 5.2). The pairwise PD plot also shows that most observations are below the 45-degree line, meaning VK usually gives lower PDs than Merton (Appendix 5.3). This is consistent with VK using a lower default point, STD + 0.5 × LTD, instead of total debt. Therefore, VK mainly makes the PD estimates more conservative, but it does not materially change the discrimination result. Overall, Merton and VK are almost the same in label-based ranking performance, and the main difference is interpretation rather than accuracy. These metrics should still be read carefully because AUC and CAP-AR test ranking, not whether the PD levels are calibrated to market prices.

6. Calibration Analysis

6.1 The CDS-implied Market Hazard (our Benchmark)

Our CDS data comes from Markit — 5-year senior unsecured spreads for 83 investment-grade U.S. firms, which gives us 313 firm-year observations from 2021 to 2025. We use Markit's 5-year default probability dp to back out a flat market hazard rate $\lambda_market = -\ln(1 - dp) / 5$. To compare models on the same scale, we also turn each model's 1-year PD into an annualised hazard $\lambda_m = -\ln(1 - PD_m)$. For Hull-White, λ_HW comes directly from the CDS PV equation (closed-form approximation $\lambda \approx s / (1 - R)$, or solved with Brent's method for the full version). For the structural models, we apply a seniority-weighted recovery rate following Altman & Kishore (1996): RR = (STD/D) × 0.55 + (LTD/D) × 0.35, reflecting that short-term debt is typically secured and long-term debt typically unsecured. The sample mean is 0.356 with a standard deviation of 0.049, ranging from 0.20 to 0.55 across firms. For Hull-White, the same firm-level RR is used in the CDS PV equation.

6.2 Task A — Does the Level Match?

The first question is whether model hazards line up with what the market prices. We compute Spearman and R^2 on the pooled panel against λ_market. Merton gives $\rho = 0.567$ with $R^2 = -2.03$; VK gives $\rho = 0.562$ with $R^2 = -2.06$. The ranking is reasonable, as both models put riskier IG firms in roughly the right order, but the absolute level is way off. A strongly negative R^2 means the model is worse than just guessing the panel mean.

To rule out the possibility that this is an artefact of the RR assumption, we additionally compare model-implied expected losses (EL = PD × LGD) against CDS par spreads directly. Par spreads represent the market's observed expected loss without requiring any RR assumption. The results are consistent: Merton $R^2 = -1.94$ and VK $R^2 = -1.95$. The level gap is a structural feature of the GBM framework, not a recovery rate problem.

Hull-White shows $\rho = 0.999$ and $R^2 = +0.99$, but this is by construction, since λ_HW is bootstrapped from the same CDS spreads we compare it to, and $EL_HW \approx parspread$ is a mathematical identity. This serves as a sanity check that our PV solver agrees with Markit pricing, not a model evaluation result. The actual takeaway from Task A is that Merton and VK need a level fix, which we attempt in 6.3.

6.3 Task A-2 — A Single Linear Calibration, Walk-Forward

If the level error is mostly a constant shift, then a simple linear rescaling $\lambda_market \approx a + b \cdot \lambda_model$ should fix most of it. We fit (a, b) using OLS on years $\leq k$, then check how it does on year $k + 1$, for three splits (2023, 2024, 2025 as the held-out year). We use an expanding window because the within-year hazard variance for IG firms is small, and a 1-year rolling window makes the slope blow up. For Merton, RMSE drops by about half across all three splits (roughly 0.009

→ 0.005), which is good — most of the level shift is really just a constant offset. But the calibrated R^2 is still negative every time: −0.51, −0.36, −0.25 across the 2023, 2024, 2025 test years. VK gives basically the same pattern. So the linear fix narrows the gap, but doesn't close it. Our guess for why: IG firms have structural λ values close to zero (the structural PD is tiny because the firms are far from default), so even after rescaling, the model can't really capture the cross-sectional spread of CDS-implied risk. Hull-White stays at +0.99 calib R^2 across all splits — again, that's by construction, not a result.

6.4 Task B—Cross-sectional Spearman vs CDS Hazard

Task B evaluates whether structural models can correctly rank firms by relative credit risk within each annual cross-section, rather than match absolute hazard levels. Model-implied hazard rates from Merton and VK/KMV are compared to CDS-implied rates using Spearman rank correlation (with Kendall as a robustness check).

Both models show moderate but economically meaningful ranking ability: average Spearman correlations over 2022–2025 are about 0.56 (Merton) and 0.55 (VK/KMV). Performance is strong in 2022, declines in 2023, and recovers thereafter, suggesting that equity-based structural signals capture relative risk but cannot fully replicate CDS market rankings.

Hull–White achieves near-perfect correlations (~0.998), but this is by construction since its hazard rates are bootstrapped from CDS pricing. It serves as a consistency benchmark rather than evidence of superior predictive power.

Overall, structural models provide useful—though imperfect—signals for risk ranking. They are suitable for credit screening, but accurate CDS pricing or level estimation requires additional calibration.

6.5 Task C—Spearman Correlation with Actual Agency Ratings

Task C introduces a second external benchmark that is independent of both the binary default label and CDS market prices. Agency ratings are converted to an ordinal scale from AAA = 1 to D = 22, and Spearman correlation is computed between each model's risk score and the actual rating. For structural models the score is negative distance-to-default; for Hull-White it is the bootstrapped hazard rate. Markit's own default probability and implied rating serve as upper-bound references.

The 2021 row contains only five firms and should be treated as a small-sample check. Hull-White shows an unusually high correlation of 0.71 that year, likely reflecting the lag between CDS market repricing post-COVID and agency rating adjustments. From 2022 to 2025, all correlations are positive and moderate. Merton ranges from 0.36 to 0.48, VK from 0.33 to 0.45, and Hull-White from 0.36 to 0.54. By 2025, Merton reached 0.48, the highest of all five proxies, while Hull-White fell to 0.36. This shows that a CDS-calibrated model is not universally superior across all validation targets.

The moderate correlations reflect a genuine difference in what each signal measures. Agency ratings are discrete and slow-moving institutional judgments, whereas CDS spreads and distance-to-default are point-in-time and sensitive to current market conditions. A firm can appear high-risk in equity or CDS data well before a rating downgrade or remain highly rated through a temporary market selloff. This timing mismatch explains why Hull-White fits CDS hazards almost perfectly in Tasks A and B but converges to the same moderate range as Merton in Task C.

Merton slightly outperforms VK in most rating-year comparisons despite VK's more refined default point, suggesting that leverage differentiation adds limited value in an investment-grade sample where firms are broadly similar in capital structure. The key takeaway is that all five proxies cluster between 0.39 and 0.43 on average, with no single winner, confirming that structural and reduced-form models capture related but distinct dimensions of credit quality.

7. Conclusion

Merton, VK/KMV, and Hull-White address the same credit-risk question from different angles, and the evidence suggests they are most useful in combination rather than in competition.

Merton and VK separate distressed firms from healthy ones with an AUC of approximately 0.957 and 0.962 respectively, confirming that distance-to-default remains a reliable cross-sectional screening tool even without CDS data. The two structural models do not, however, align with CDS-implied default levels — both show negative R-squared values against market hazard rates, and affine calibration narrows but does not close the gap. This is not a failure of parameterization; it reflects the structural models' shared GBM assumption and their inability to generate the short-end hazard rates that CDS markets price in.

Hull-White is a different kind of tool. Because it bootstraps default intensity directly from observed 5-year CDS spreads, it naturally fits the CDS market — Task A R-squared exceeds 0.99 and the Spearman correlation with market-implied hazard is approximately 0.999. This does not make it universally superior. Its coverage is limited to firms with liquid CDS quotes, which systematically excludes the most distressed names that are most relevant for default prediction. It also offers no ranking advantage over Merton when evaluated against an independent benchmark: Task C shows all five proxies clustering between 0.39 and 0.43 against actual agency ratings, with no single winner.

Task C is the more informative test precisely because it is independent of both the binary default label and the CDS market price. The moderate positive correlations confirm that each model captures some dimension of credit quality, while the absence of a clear winner reflects a genuine truth — agency ratings, CDS spreads, and equity-based structural signals measure related but distinct aspects of credit risk. A practical workflow should therefore use the models in layers: Merton and VK for broad firm-level screening where CDS data is unavailable; Hull-White for names with liquid CDS where precise hazard calibration is needed; and agency ratings as a slow-

moving institutional benchmark against which both can be periodically validated. Future work could extend the CDS bootstrap to multiple tenors, test ranking stability against subsequent realized downgrades, and examine the sensitivity of results to alternative recovery rate specifications beyond the seniority-weighted approach adopted here.

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Appendix

Appendix 2.1.1 Merton Top-Risk Table

Data Source / Category	Variable	Source Field / Series	Processing
Yahoo Finance — Equity Market Data	Daily price P_t	Adj Close	Adjusted close price
Yahoo Finance — Equity Market Data	Market cap E	price × sharesOutstanding	Year-end close × shares outstanding
Yahoo Finance — Equity Market Data	Equity volatility σ_E	OptionMetrics implied vol (primary) / Daily log returns (fallback)	See Section 1.2 above
Yahoo Finance — Annual Balance Sheet	Total debt D	Total Debt	Direct; fallback: STD + LTD
Yahoo Finance — Annual Balance Sheet	Short-term debt STD	Current Debt	Direct
Yahoo Finance — Annual Balance Sheet	Long-term debt LTD	Long Term Debt	Direct
Yahoo Finance — Annual Balance Sheet	Default point DP	Computed	DP = STD + 0.5 × LTD
Yahoo Finance — Annual Balance Sheet	Total assets A	Total Assets	Direct
FRED — Risk-Free Rate	Risk-free rate r	DGS1 (1Y Treasury CMR)	Annual average

Appendix 2.2.1 Filtering Criteria

From the raw option data (millions of rows per year), we apply the following filters to extract a single ATM implied volatility per firm per year:

Step	**Filter**	**Rationale**
1. Ticker	Keep only our 163 sample firms	Match the equity/balance-sheet panel
2. Observation date	Last 5 trading days of December each year	Align with year-end equity and debt data

3. Option type	Call only (cp_flag = 'C')	Calls are typically more liquid at ATM; put-call parity implies identical IV at the same strike
4. Days to expiry	250 – 450 calendar days	Target ≈ 1 year to match the model horizon T = 1; range allows flexibility when no exact 365-day option exists
5. Moneyness (ATM)	Delta ∈ [0.40, 0.60]	ATM options have the highest vega (sensitivity to volatility) and the tightest bid-ask spreads, yielding the most informative IV estimate. Delta ≈ 0.50 corresponds to at-the-money under Black-Scholes
6. IV quality	0 < impl_volatility < 3.0	Remove missing values and extreme outliers (>300% annualised vol)
7. Liquidity	volume > 0 or open_interest > 0	Ensure the option has actual market activity; zero-activity options may have stale or model-based IV
8. Selection	Per ticker-year, pick the row with delta closest to 0.50; break ties by days closest to 365	One observation per firm-year; prioritise the most at-the-money, closest-to-target-tenor option

Appendix 2.2.2 Coverage

Year	Tickers with IV	Missing	Notes
2021	160 / 163	ITW, RSG, SJM	No qualifying options at year-end
2022	158 / 163	ITW, KMI, RSG, T, TRV	No qualifying options at year-end
2023	162 / 163	RSG	No qualifying options at year-end
2024	162 / 163	FOX	No qualifying options at year-end
2025	146 / 163	17 tickers (mostly distressed)	OptionMetrics data available through Aug 2025; many distressed firms delisted

For firm-years where implied volatility is unavailable, σ_E falls back to the historical volatility estimate.

Appendix 2.3.1 Default Label Definition

Category	Criteria	Firms
Formal bankruptcy	Court-ordered Chapter 11 filing	HTZ (May 2020)
Going-concern warning	SEC/auditor going-concern opinion	PLUG, FCEL (2023)
Severe financial distress	Persistent net losses, unsustainable cash burn, negative equity, or high leverage (at least one criterion met)	LCID, RIVN, SPCE, WKHS, QS, BLNK, AMC, BYND, TLRY, CLOV, PTON, W, OPEN, SKLZ, LMND, RIG, CLF, AAL

Limitation: The labelling standard is heterogeneous — HTZ experienced actual default, while the rest exhibit distress characteristics. The models may be learning "distress features" rather than strict default probability. This should be kept in mind when interpreting results.

Appendix 4.1.1 Merton Top-Risk Table

Sorted by merton_PD descending. Default label: 1 = distressed/default, 0 = non-default.

Ticker	Year	Default	Merton DD	Merton PD
BYND	2025	1	−0.302	61.86%
SKLZ	2022	1	0.388	34.90%
OPEN	2022	1	0.426	33.52%
WKHS	2025	1	0.514	30.36%
SPCE	2024	0	0.566	28.56%
OPEN	2024	1	0.662	25.39%
OPEN	2025	1	0.668	25.20%
WKHS	2024	1	0.689	24.55%
HTZ	2024	1	0.717	23.68%
AMC	2024	1	0.725	23.43%
BYND	2024	1	0.845	19.91%

AMC	2023	1	0.878	18.99%
FCEL	2024	1	0.911	18.12%
TLRY	2025	1	0.936	17.46%
SKLZ	2023	1	0.956	16.96%

Appendix 4.2.1. Top 15 Firms by VK/KMV Probability of Default

Sorted by vk_PD descending. Default point DP = STD + 0.5 × LTD.

Ticker	Year	Default	VK DD	VK PD
BYND	2025	1	−0.032	51.29%
OPEN	2022	1	0.551	29.08%
SKLZ	2022	1	0.574	28.28%
WKHS	2025	1	0.617	26.86%
WKHS	2024	1	0.703	24.10%
OPEN	2024	1	0.851	19.73%
SPCE	2024	0	0.853	19.69%
HTZ	2024	1	0.910	18.13%
OPEN	2025	1	0.924	17.78%
AMC	2024	1	1.108	13.39%
SKLZ	2023	1	1.175	11.99%
BYND	2024	1	1.188	11.74%
FCEL	2024	1	1.248	10.60%
HTZ	2025	1	1.318	9.38%
AMC	2023	1	1.335	9.09%

Appendix 4.3.1. Hull-White Hazard Rate Summary Statistics

398 firm-year observations, 84 firms, 2021–2025. RR = seniority-weighted (Altman & Kishore, 1996).

	λ (triangle)	λ (PV)	PD 1Y	PD 5Y	Rec. Rate
Count	398	398	398	398	398
Mean	0.008862	0.008818	0.008759	0.042651	0.3653
Std	0.006537	0.006505	0.006356	0.029068	0.0472
Min	0.002602	0.002589	0.002586	0.012863	0.2000
25%	0.005459	0.005432	0.005417	0.026794	0.3499
50%	0.006985	0.006950	0.006926	0.034152	0.3647
75%	0.010037	0.009987	0.009937	0.048709	0.4000
Max	0.064784	0.064465	0.062431	0.275536	0.5500

Appendix 4.3.3. Sanity Check: HW Bootstrap vs Markit 5Y Default Probability

Comparison of hw_PD_5y against Markit dp across 313 matched firm-year observations.

Metric	Value
Mean residual (HW − Markit)	−0.003254
Std residual	0.005021
Spearman rank correlation	0.9817

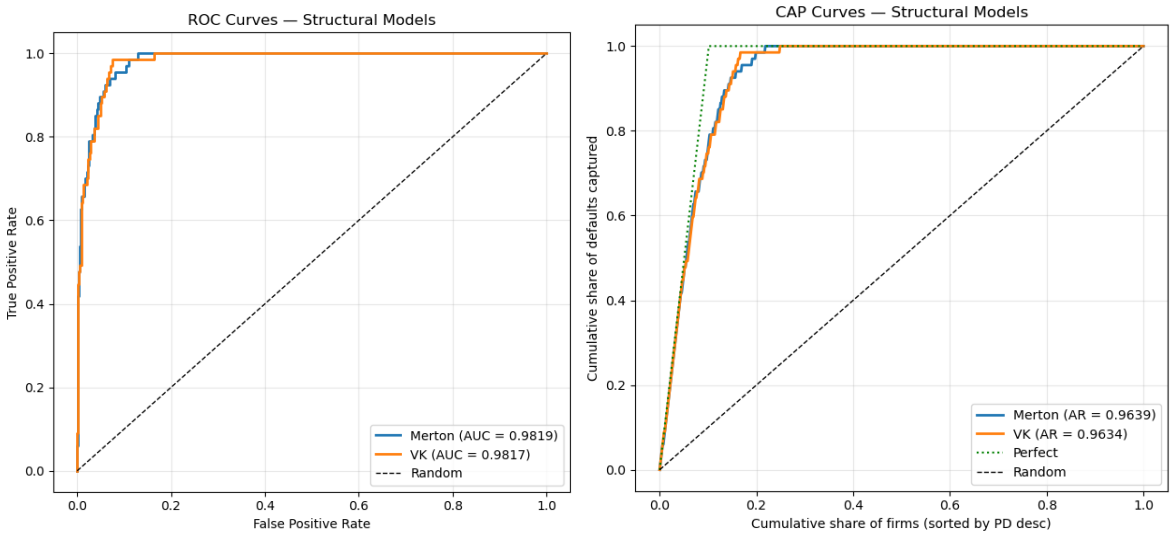
Appendix 4.3.2. Top 15 Firms by Hull-White CDS-Bootstrap Hazard

Sorted by hw_lambda descending. Spread in basis points.

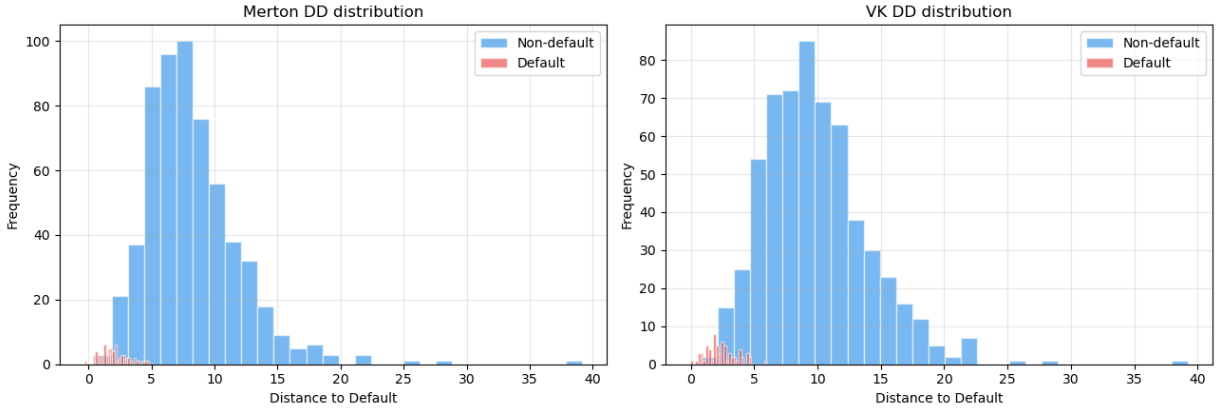
Ticker	Year	Rating	Spread (bps)	λ (HW)	PD 5Y
F	2022	BB	377.3	0.0645	27.55%
MU	2021	BBB	281.0	0.0621	26.71%
DAL	2022	BB	379.5	0.0524	23.03%
DAL	2021	BB	224.2	0.0372	16.96%
F	2023	BBB	206.4	0.0349	16.01%

OXY	2021	BB	176.3	0.0292	13.60%
F	2025	BBB	160.1	0.0272	12.71%
F	2024	BBB	156.5	0.0265	12.43%
DAL	2023	BB	180.8	0.0250	11.73%
F	2021	BB	147.8	0.0245	11.53%
BA	2022	BBB	140.4	0.0220	10.42%
LEN	2022	BBB	141.6	0.0217	10.27%
NFLX	2022	BBB	140.9	0.0216	10.22%
OXY	2022	BB	146.2	0.0215	10.18%
SYY	2021	BBB	125.9	0.0209	9.91%

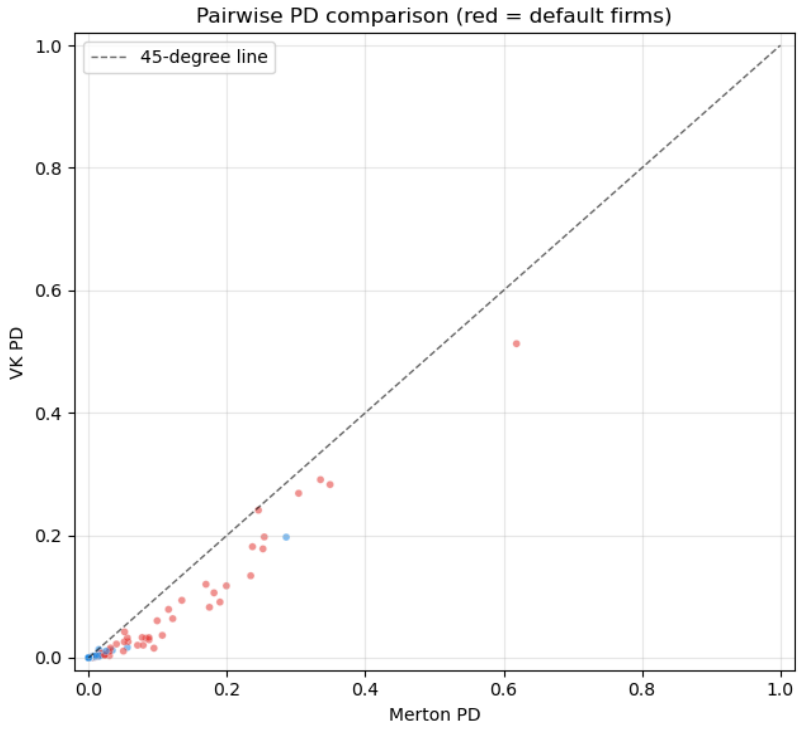
Appendix 5.1 ROC & CAP Curve



Appendix 5.2 DD Distribution



Appendix 5.3 Pairwise PD comparison



Appendix 6.1. Pooled-Panel Level Matching against the CDS-Implied Market Hazard

Pooled panel, 313 firm-year observations. λ_market derived from Markit 5Y dp. Spearman ρ and R² computed against λ_market.

Model	n	Spearman ρ	Raw R²	RMSE
Merton	313	0.567	−2.03	0.0117
VK / KMV	313	0.562	−2.06	0.0117
Hull-White (CDS bootstrap)	313	0.999	+0.99	0.0006

Note. *Hull-White λ is bootstrapped from the same 5Y CDS spread that λ_market is derived from, so the two are essentially the same number through different formulas. We report HW here as a sanity check on our PV implementation, not as a separate predictive model.*

Appendix 6.2. Walk-Forward Linear Calibration

OLS fit λ_market ≈ a + b·λ_model on expanding window ≤ split year; evaluated on hold-out year. Raw R² uses the un-calibrated model; Calib R² uses the fitted (a, b).

Model	Split	n_train	n_test	a	b	Raw R²	Calib R²
Merton	≤ 2022 → 2023	77	78	0.0130	0.398	−3.06	−0.51
Merton	≤ 2023 → 2024	155	79	0.0111	0.470	−4.02	−0.36
Merton	≤ 2024 → 2025	234	79	0.0102	0.517	−3.99	−0.25
VK	≤ 2022 → 2023	77	78	0.0130	1.207	−3.06	−0.50
VK	≤ 2023 → 2024	155	79	0.0111	1.401	−4.08	−0.37
VK	≤ 2024 → 2025	234	79	0.0102	1.516	−3.99	−0.26
HW	≤ 2022 → 2023	77	78	0.0006	0.975	+0.99	+0.99
HW	≤ 2023 → 2024	155	79	0.0004	0.990	+0.99	+0.99
HW	≤ 2024 → 2025	234	79	0.0003	0.995	+0.99	+0.99

Note. *We tried a 1-year rolling window first, but it did not work — Merton's slope shot up above 80 in the 2024 split because IG firms have almost no within-year hazard variance. Expanding windows is what we end up using everywhere. The HW rows are included for completeness, but as in Appendix 6.1 they are essentially fixed by construction (the OLS slope is ≈ 1 because the predictor and target are the same CDS quantity).*